Integrative Group Work 1
PMX006: Bictegravir

Authors:

Daan Demeulenaere,

Shanice Sokeng Tsake,

Seyed Mostafa Fazel Hosein

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1. Bibliographic and Contextual Work

1.1 Clinical Pharmacology of Bictegravir

Bictegravir (ATC code: J05AR20) is an integrase strand transfer inhibitor (INSTI) used in HIV treatment. It inhibits HIV integrase, preventing viral DNA integration into the host genome (1, 2). Although bictegravir exhibits potent antiviral activity on its own, it is administered as a fixed-dose combination with emtricitabine and tenofovir alafenamide, both nucleos(t)ide reverse transcriptase inhibitors (NRTIs) (Biktarvy) (1, 2). Combination antiretroviral therapy (cART) is essential, as HIV can rapidly develop resistance when a single antiretroviral drug is used alone, due to the virus's ability to mutate and generate resistant variants (3). By targeting different stages of viral replication, cART can suppress viral replication to undetectable levels, preserve immune function and prevent disease progression, but does not eradicate HIV from the body (3, 4).

Bictegravir has a high genetic barrier to resistance. No treatment-emergent resistance was identified in 11 patients analysed through 144 weeks, although *in vitro* mutations M50I and R263K reduced susceptibility by 1.3-fold and 2.2-fold, respectively (1).

Biktarvy effectively suppresses HIV replication. At week 48, 89–92% of previously untreated adults achieved HIV-1 RNA levels below 50 copies/mL, while viral suppression below this threshold was maintained in 98% (49/50) of adolescents who switched to Biktarvy (1). Bictegravir alone exhibits potent antiviral activity against HIV-1 (EC50: <0.05–6.6 nM) and HIV-2. Its protein-adjusted EC95 against wild-type HIV-1 is 361 nM (0.162 µg/mL) (1, 2). A trough concentration of 760 ng/mL (inhibitory quotient = 4.7) has been proposed as a reference threshold for adequate virological suppression in Phase III trials, although no clinically validated exposure target has been established (2). Similarly, no clinically validated upper exposure limit has been established. Bictegravir has a generally favourable safety profile, with the most frequently reported adverse effects being headache (5%), diarrhoea (5%) and nausea (4%) (1).

Bictegravir is classified as a Biopharmaceutics Classification System (BCS) class II drug, characterised by low solubility and high permeability (2, 5). Its estimated oral bioavailability exceeds 61%, with dose linearity reported between 25 and 100 mg. Peak plasma concentrations are reached within 2–4 hours, and a moderate- or high-fat meal increases AUC by approximately 24% compared with the fasted state, without a clinically relevant effect (2). Bictegravir exhibits extensive plasma protein binding (>99%), with an unbound fraction of approximately 0.25% in vitro. It is primarily metabolised by CYP3A and UGT1A1 (2). Approximately 60.5% of a radiolabelled dose is recovered in faeces and 35% in urine, predominantly as metabolites; only 1.3% is excreted unchanged in urine (2). Renal excretion of unchanged bictegravir is therefore a minor elimination pathway. Bictegravir is also a substrate of P-glycoprotein (P-gp) and breast cancer resistance protein (BCRP) and has a plasma elimination half-life of approximately 17.3 hours (2).

1.2 Drug Development Context

The dataset originates from a Phase 1a single ascending dose (SAD) study investigating Gajeevy, an investigational drug with structural and mechanistic similarities to bictegravir. The study was designed as a four-arm, parallel-group study with oral administration. A total of 32 healthy volunteers received single doses ranging from 10 to 100 mg. The primary objectives were to assess safety and tolerability and, where applicable, explore the maximum tolerated dose (6). Secondary objectives included characterising the pharmacokinetic and pharmacodynamic properties of Gajeevy (6). The findings will inform dose selection for Phase II trials, which investigate preliminary efficacy, dose–response relationships and safety in patients (6). Phase III trials subsequently evaluate the drug's safety and efficacy in larger patient populations to confirm its benefit–risk profile before regulatory approval and market authorisation (6).

The main objective of this research was to clean and explore the dataset, followed by an exploratory pharmacokinetic analysis of the drug.

1.3 Pharmacokinetic Challenges

- The co-administration of bictegravir with divalent cations (e.g., magnesium, iron, calcium and zinc) can significantly reduce its bioavailability (2).
- As a result of co-administration with potent dual inhibitors or inducers of CYP3A4 and UGT1A1 (e.g., atazanavir, a protease inhibitor), clinically significant drug-drug interactions (DDI) are expected (2).
- Pregnancy alters bictegravir pharmacokinetics. Despite evidence supporting its use during pregnancy, data on safety during breastfeeding and effects on infants remain limited (1, 7).
- Bictegravir pharmacokinetics in adults aged 65 years and older remain insufficiently characterised, despite no clinically relevant age effect on exposure being identified in population pharmacokinetic analyses (1).

1.4 Pharmacodynamic Challenges

- As mentioned above, no clinically validated exposure target has been established due to the absence of a clear exposure–response relationship. The proposed threshold of 760 ng/mL is based on theoretical considerations and clinical observations (2).
- Suboptimal adherence may lead to inadequate drug exposure and ongoing viral replication, potentially promoting the selection of resistant HIV variants. This may reduce bictegravir susceptibility and compromise long-term virological suppression. Therefore, maintaining treatment adherence is essential to preserve treatment efficacy (1, 3).

2. Data Handling and Management

The dataset provided was already cleaned. Nevertheless, all variables and patient records were systematically assessed for missing values, potential outliers, and inconsistencies using R. Descriptive statistics, distribution plots, individual concentration–time profiles, and patient-level consistency checks were performed. No additional anomalies requiring correction were identified. Finally, dose and concentration units were converted, and the dataset was prepared for subsequent pharmacokinetic analyses.

3. Pharmacokinetic Analysis

3.1 Non-compartmental analysis

A non-compartmental analysis (NCA) was performed on the clean dataset using PKAnalix2024R1. The NCA methodology, including parameter calculation and acceptance criteria, is described in the reproducibility document. The median Tmax of 2.00 h [1.50–2.00] indicates relatively rapid absorption. The mean Cmax (4.09 ± 3.18 µg/mL) and AUC0–inf (117.18 ± 101.41 h·µg/mL) showed substantial variability which may partly reflect the different administered doses and interindividual differences. Gajeevy exhibited a mean terminal half-life of 20.12 ± 6.51 h, indicating relatively prolonged terminal elimination and suggesting that once-daily administration warrants further investigation. The mean apparent clearance (CL/F) was 436.38 ± 109.48 mL/h, while the apparent volume of distribution (Vd/F) was 11.89 ± 2.52 L. However, both parameters remain dependent on the unknown oral bioavailability.

Comparison with single-dose bictegravir data from the first-in-human study (see reproducibility document) showed a similar absorption and elimination profile for Gajeevy at 25–100 mg (median Tmax: 1.75–2.00 h vs. 2.00–3.00 h; terminal half-life: 16.9–20.4 h vs. 16.7–18.9 h), while Gajeevy exhibited higher exposure (Cmax 1.2–1.3-fold and AUC0–inf 1.4–1.6-fold that of bictegravir, suggesting lower apparent clearance) and consistent dose proportionality (2.21–2.64 vs. 1.43–1.63 µg·h/mL per mg) (8). Overall, while these findings support the pharmacokinetic plausibility of the NCA results, they do not constitute formal validation due to differences in study populations, sample sizes, summary statistics (medians for Gajeevy vs. means for bictegravir Cmax and AUC0–inf), and unknown prandial conditions for the Gajeevy subjects compared to the fasted state of the bictegravir reference study (8).

3.2 Exploratory Covariate Analysis

Relationships between seven NCA-derived PK parameters (Cmax, Tmax, $AUC_{0-\infty}$, elimination half-life ($T_{1/2}$), elimination rate constant (k_e), CL/F, and Vd/F) and five continuous covariates (age, body weight, albumin, serum creatinine, and CRP) were evaluated using Spearman correlation. Categorical covariates, including sex and genetic polymorphisms (ABCB1, ABCG2, CYP3A5, and UGT1A1), were explored using boxplots. Of 35 correlations, based on Table S1, five had unadjusted p-values below 0.05, but only two remained significant

after Benjamini–Hochberg correction: body weight with Vd/F ($\rho = 0.55$, adjusted $p = 0.0377$) and age with CL/F ($\rho = -0.52$, adjusted $p = 0.0405$). These findings suggest that body weight and age may contribute to PK variability, although their impact on clinical efficacy was not assessed. Plots and tables are mentioned in reproducibility file.

3.3 Compartmental analysis

A compartmental analysis (CA) was performed on our clean dataset in PKAnalix2024R1.

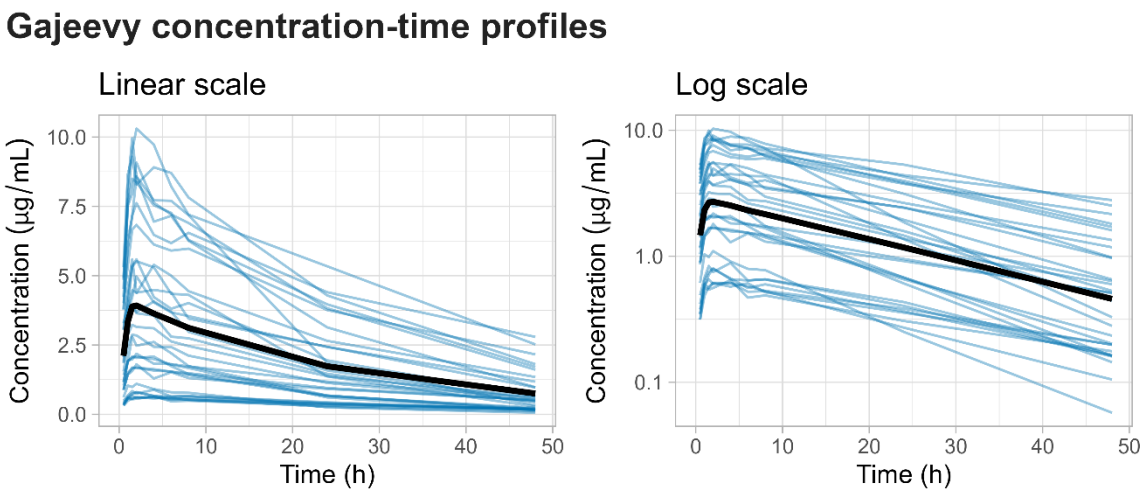


Figure 1.1: Concentration-time profiles following the oral administration of Gajeevy on both linear and log scale.

NCA provided an initial overview of the pharmacokinetic characteristics of Gajeevy following oral administration, without assuming a specific compartmental model. The observed concentration–time profiles (Figure 1.1), together with the NCA-derived parameters (Section 3.1), were subsequently used to guide the selection of a preliminary compartmental model. The concentration–time profile supports the selection of an oral one-compartment model with first-order absorption and elimination. A characteristic oral absorption pattern is observed, with concentrations increasing exponentially from the first sampling point at 0.5 hours to a peak concentration at approximately 2 hours, followed by a gradual decline (9). This resembles a linear absorption process from the gastrointestinal tract into the systemic circulation, as no clear plateau is observed before Tmax and Cmax (9). Subsequently AUC appears to increase approximately proportionally with dose (Figure 1.2; Section 3.1). These observations suggest dose-proportional pharmacokinetics and support the use of a one-compartment model with first-order absorption (9). As no data were available before the first sampling point, no absorption lag time was included, keeping the model as simple as possible without introducing an additional parameter unsupported by the available data. The post-peak decline appears approximately linear on the semi-logarithmic scale from 2 to 48 hours, which is consistent with first-order elimination. Finally, the absence of an obvious inflection point indicating a distinct distribution phase supports the use of a one-compartment model as a reasonable starting point.

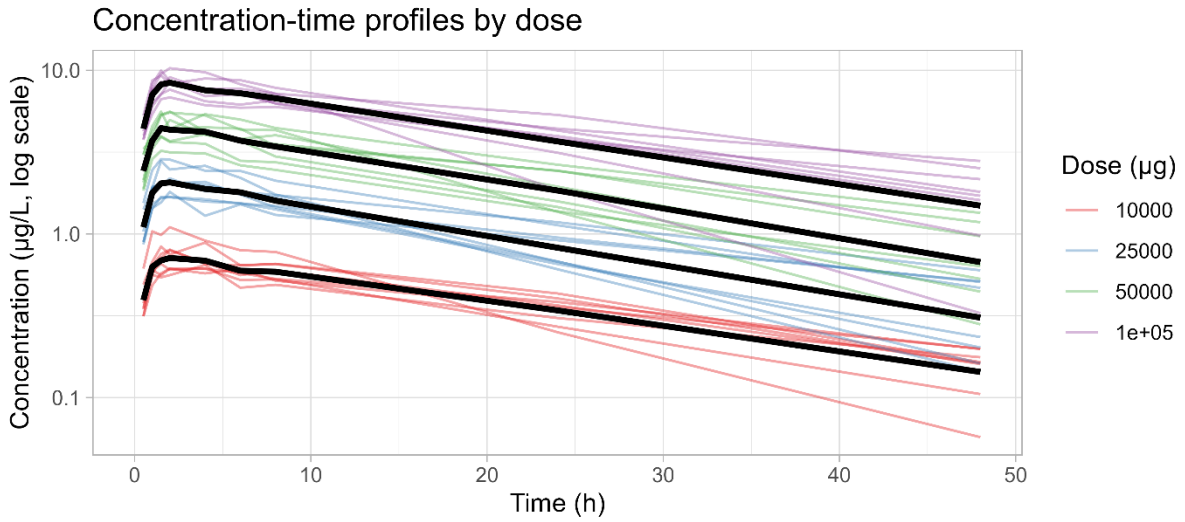


Figure 1.2: Concentration-time profiles separated on dose following the oral administration of Gajeevy on log scale.

However, when we look at the concentration-time curves per dose, we see a slight bend (inflection point) between 4 and 8 hours after dosing in all dose groups. This could indicate a two-compartment system, in which the drug distributes into a second compartment (9). When we fit this model to our data, the two-compartment model yields a lower -2LL (-532.8 versus -442.7), and thus a better raw fit to the data (Table 2) (9). This is consistent with the fact that models with more parameters can always follow the data at least as closely, so -2LL rewards additional parameters, even when these mainly capture noise. Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC), however, also correct for model complexity by penalizing each additional parameter (AIC = -2LL + 2 × number of parameters; BIC = -2LL + ln(n) × number of parameters), so that a model only wins if it improves the fit by more than the penalty costs (9). Because each individual was fitted separately, the two-compartment model has 64 more parameters than the one-compartment model (161 versus 97), and this additional complexity is not justified by the gain in fit: both the AIC (-248.7 versus -210.8) and the BIC (-226.1 versus -175.6) are lower for the one-compartment model (Table 2). The slight bend observed between 4 and 8 hours may reflect variability or noise in the data rather than a true distribution phase. The two-compartment model, being more flexible, can follow this deviation, but the data do not support a distinct second compartment. The preliminary one-compartment model is therefore the simpler and more reliable choice, and it describes the data adequately. This was further supported by the observed-versus-predicted plot (Figure S1), which showed no clear curvature or trend indicating systematic over- or underestimation of concentrations. A schematic representation of the model structure is shown in Figure S2.

Table 2: Fit criteria for one- and two-compartment models.

Parameter/ Criteria	1-Compartment Model	2-Compartment Model
Total Cost	1.16	0.85
-2 Log-Likelihood (-2LL)	-442.72	-532.77
AIC	-248.72	-210.77
BIC	-226.13	-175.55

4. Conclusion and Perspectives

This Phase Ia SAD study provided an initial pharmacokinetic characterisation of Gajeevy following oral administration of 10, 25, 50 and 100 mg in 32 healthy volunteers. Non-compartmental analysis, exploratory covariate assessment and compartmental modelling provided a quantitative foundation for further drug development.

NCA revealed a median Tmax of 2 h and a mean terminal half-life of 20.12 ± 6.51 h, suggesting relatively rapid absorption and prolonged plasma persistence. Comparison with single-dose bictegravir data (25–100 mg) showed a similar absorption and elimination profile, with a median Tmax of 1.75–2.00 h versus 2.00–3.00 h and a terminal half-life of 16.9–20.4 h versus 16.7–18.9 h (8). However, Gajeevy exhibited higher overall exposure, with Cmax 1.2–1.3-fold and AUC0–inf 1.4–1.6-fold higher than bictegravir, suggesting lower apparent clearance (8). Furthermore, dose-normalised AUC values remained stable (2.21–2.64 vs. 1.43–1.63 µg·h/mL per mg), indicating consistent dose proportionality (8). While these findings support the pharmacokinetic plausibility of the NCA results, they do not constitute formal validation due to differences in study populations, sample sizes, summary statistics, and unknown prandial conditions compared to the fasted bictegravir reference study (8).

Exploratory covariate analyses identified associations between age and apparent oral clearance ($\rho = -0.52$, $p = 0.0023$) and between body weight and apparent volume of distribution ($\rho = 0.55$, $p = 0.0011$). These findings suggest potential sources of pharmacokinetic variability but require confirmation in larger, dose-adjusted analyses, including evaluation of their clinical impact.

Among the compartmental models evaluated, the one-compartment model with first-order absorption and elimination was provisionally preferred based on information criteria. However, this represents a preliminary model selection rather than definitive validation of Gajeevy's distribution characteristics.

Although the sample size was appropriate for an exploratory Phase Ia study, the limited number of participants per dose cohort restricts the robustness of covariate analyses. Moreover, the single-dose design prevents assessment of accumulation and steady-state exposure, while the absence of pharmacodynamic and safety outcomes in the analysed dataset limits conclusions regarding antiviral efficacy, tolerability and optimal dosing.

Future studies should prioritise formal dose-proportionality assessment, validation of compartmental models and confirmation of potential covariate effects. Multiple-ascending-dose studies will be essential to investigate accumulation, steady-state pharmacokinetics and the feasibility of once-daily administration. Subsequent PK/PD analyses should establish exposure–response relationships to support evidence-based dose selection.

Overall, the findings provide a coherent preliminary pharmacokinetic profile of Gajeevy, supported by exploratory comparisons with bictegravir, while highlighting the additional evidence required for its continued clinical development.

5. References

1. European Medicines Agency. Biktarvy: product information [Internet]. Amsterdam: European Medicines Agency; [cited 2026 Oct 23]. Available from: https://www.ema.europa.eu/en/medicines/human/EPAR/biktarvy

2. Ekobena P, Briki M, Dao K, Marzolini C, Andre P, Buclin T, et al. Population pharmacokinetics of bictegravir in real-world people with HIV. J Antimicrob Chemother. 2025;80(10):2782–2789. doi:10.1093/jac/dkaf297.

3. Gilead Sciences, Inc. About HIV treatment [Internet]. Foster City (CA): Gilead Sciences, Inc.; [cited 2026 Oct 23]. Available from: https://www.biktarvy.com/what-is-hiv/about-hiv-treatment

4. Gilead Sciences, Inc. What is HIV? [Internet]. Foster City (CA): Gilead Sciences, Inc.; [cited 2026 Oct 23]. Available from: https://www.biktarvy.com/what-is-hiv

5. Subramanian R, Wang J, Murray B, Custodio J, Hao J, Lazerwith S, et al. Human pharmacokinetics prediction with an in vitro–in vivo correction factor approach and in vitro drug-drug interaction profile of bictegravir, a potent integrase-strand transfer inhibitor component in approved Biktarvy® for the treatment of HIV-1 infection. Xenobiotica. 2023;53(11):1020–1030. doi:10.1080/00498254.2023.2169207.

6. Kandi V, Vadakedath S. Clinical trials and clinical research: a comprehensive review. Cureus. 2023;15(2). doi:10.7759/cureus.35077.

7. Holt LM, Short WR, Momplaisir F, Hyun E, McKinney J, Lugo Morales A, et al. Bictegravir use during pregnancy: a multicenter retrospective analysis evaluating HIV viral suppression and perinatal outcomes. Clin Infect Dis. 2024;79(5):1258–1261. doi:10.1093/cid/ciae218.

8. Gilead Sciences, Inc. Study GS-US-141-1218: bictegravir pharmacokinetic parameters following single doses in healthy subjects (Part A), Table 1-1. Clinical study protocol, Amendment 2, NCT03547908. Foster City (CA): Gilead Sciences, Inc.; 2026. Available from: https://cdn.clinicaltrials.gov/large-docs/08/NCT03547908/Prot_000.pdf [accessed 2026 Oct 26].

9. Mould DR, Upton RN. Basic concepts in population modeling, simulation, and model-based drug development—Part 2: introduction to pharmacokinetic modeling methods. CPT Pharmacometrics Syst Pharmacol. 2013;2(4). doi:10.1038/psp.2013.14.

6. Supplementary material

Table S1: Summary of covariate analysis

Pk-parameter	Covariate	N	Rho	P-value	BH-adjusted p
Vd/F (mL)	Body weight (kg)	32	0.55	0.00108	0.0377
CL/F (mL/h)	Age (years)	32	-0.52	0.00232	0.0405
kel (1/h)	Body weight (kg)	32	-0.41	0.01930	0.2010
T1/2 (h)	Body weight (kg)	32	0.40	0.02300	0.2010
T1/2 (h)	Age (years)	32	0.35	0.04690	0.2980

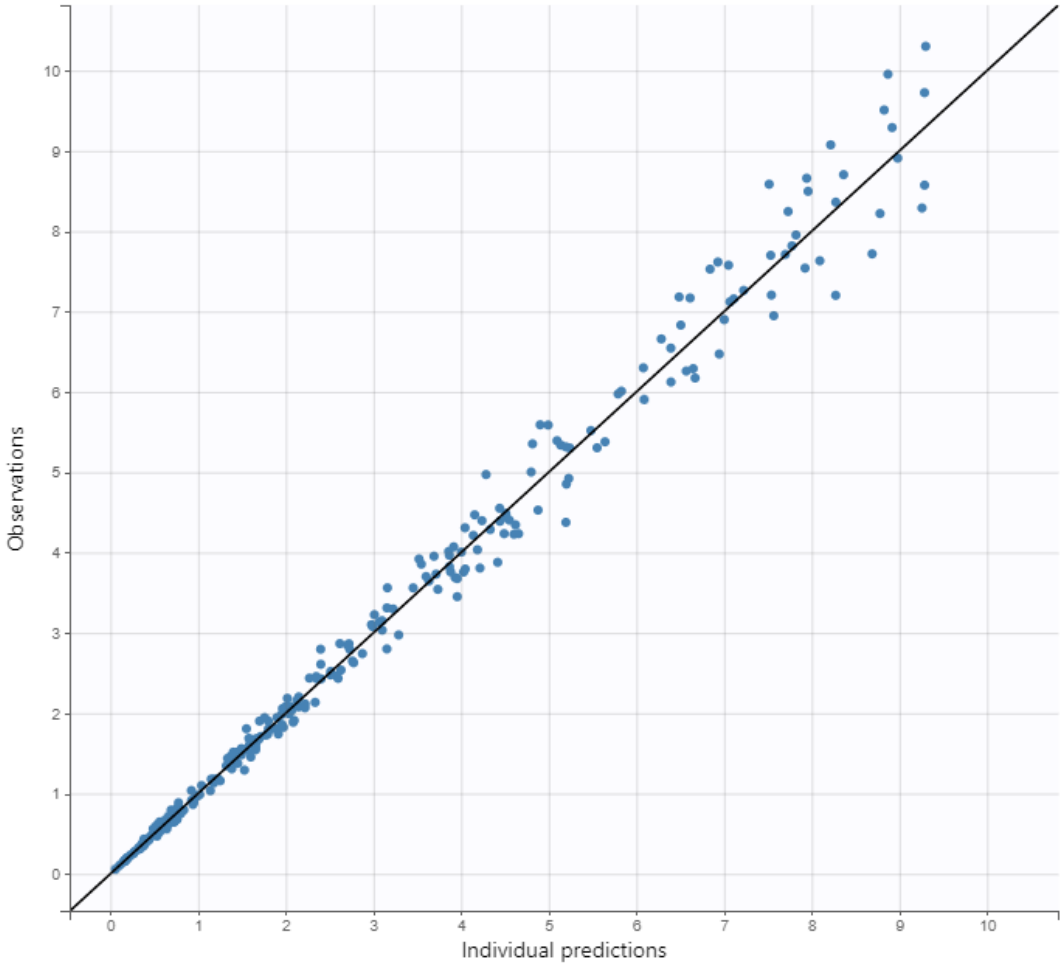


Figure S1: Observed versus individual predictions of concentration (µg/mL)

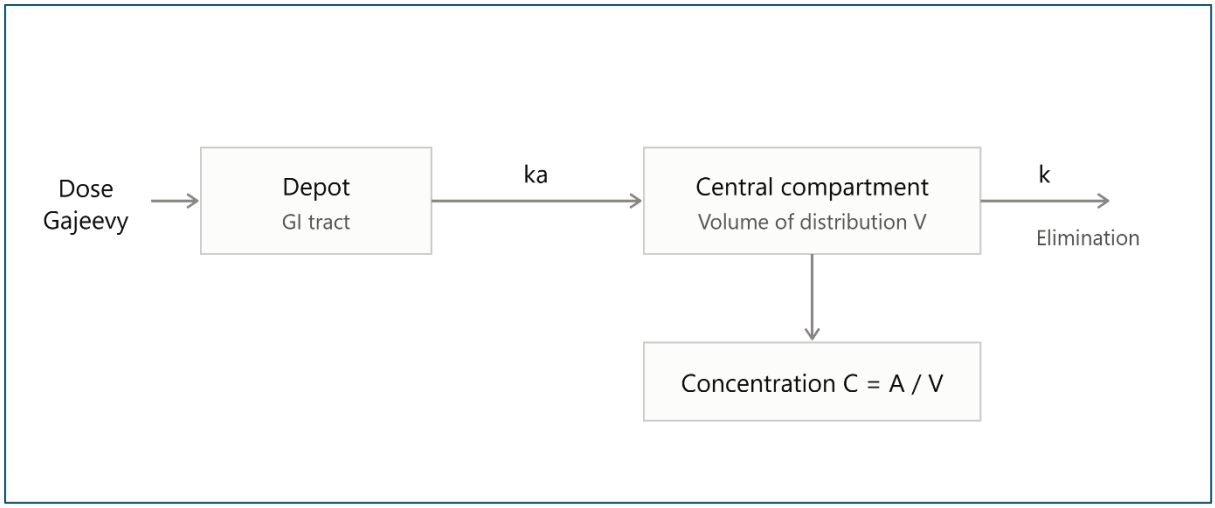


Figure S2: Schematic representation of the model structure of oral Gajeevy administration